

Market Efficiency and Externalities

Martin J. Conyon

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A companion to the lecture slides. Each slide that hides a step is reproduced here and worked through line by line.

Before we begin

Required readings

Required readings

The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 7** — Consumers, Producers, and the Efficiency of Markets.
- **Chapter 10** — Externalities.

Two chapters. Read both alongside these slides.

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Slide 2

Working through it

Why two chapters. These slides covers two distinct pieces of Mankiw. Chapter 7 builds the measuring instrument — consumer surplus, producer surplus, total surplus — and uses it to show that a competitive market maximizes the gains of its participants. Chapter 10 then breaks that result by letting a trade affect somebody who is not party to it.

What the slides add. Two things. First, the demand curve is *built* here from individual willingness to pay rather than assumed, so that every surplus triangle can be traced back

to named buyers. Second, deadweight loss is defined in its general form — surplus lost by trading the wrong quantity, in *either* direction — because the externality half of the session needs the too-much case and the too-little case equally.

How to use them. Read chapter 7 before the material and chapter 10 after. The efficiency result in chapter 7 is the benchmark; chapter 10 only makes sense as a departure from it.

Where we are going

Where we are going

1. **What a buyer will pay** — value becomes a number.
2. **From one buyer to the market** — adding demands sideways.
3. **Consumer surplus** — the buyers' gain, measured.
4. **Producer surplus** — the sellers' gain, measured.
5. **Is the market any good?** — the free market gets its grade.
6. **Costs that spill over** — smoke, bystanders, and too much trade.
7. **Benefits that spill over** — the mirror image: too little.
8. **Fixing it** — taxes, permits, and bargains with no government at all.

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Slide 3

Working through it

The shape of the argument. Eight parts, and they divide cleanly in two.

Parts 1–5 build an instrument and then use it. Parts 1 and 2 turn value into a number and assemble individual demands into a market demand. Parts 3 and 4 measure what buyers and sellers each gain. Part 5 adds the two gains and asks whether any rearrangement could do better. The answer is no, and that is the strongest result in the session.

Parts 6–8 break it. The efficiency verdict rests on an assumption that everybody affected by a trade is party to it. Part 6 drops a cost on a bystander, Part 7 drops a benefit, and in both cases the market trades the wrong quantity. Part 8 is the toolkit for putting it right.

One thread. Every result is an area on a supply-and-demand diagram. Learn to see the wedge between the two curves and most of what follows is already visible.

1. Part 1: What a Buyer Will Pay

Grading the market

Grading the market

A market left alone settles where supply meets demand.

Question one: how much good does that do for the people trading in it? Not a feeling — a dollar figure.

Question two: what about people *outside* the trade — the neighbor breathing the smoke, the stranger spared the traffic?

First half: build the measuring instrument. Second half: watch it catch the market failing.

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Slide 5

Working through it

What this frame is for. It is not teaching a technique. It is naming the two questions the rest of the session answers, so that each piece of machinery has a visible purpose.

Question one: how much good does a market do? We have predicted where a market settles. We have never asked whether settling there is worth anything. Answering that requires a measurement in dollars, not an opinion — and building it is the business of Parts 1 to 5.

Question two: what about everyone else? Every trade so far has been treated as a private matter between two people. Many are not. The neighbor breathing wood smoke and the driver spared a traffic jam are affected by trades they never agreed to, and Parts 6 to 8 ask what that does to the verdict.

Why the order matters. The second question cannot even be posed until the first is answered. “The market trades too much” is meaningless without a standard for what the right amount would be, and that standard is total surplus.

Value becomes a number

Value becomes a number

Willingness to pay (WTP). The maximum a buyer would hand over for a unit of a good. One cent above it, the buyer walks away.

WTP converts “how much do you want it” into a dollar figure — one per buyer, per unit.

Value is not a mood. It is the price at which you stop.

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Slide 6

Working through it

The definition. A buyer’s **willingness to pay** is the maximum they would hand over for one unit. One cent above it, they walk away.

Why it is the right definition. It converts something that sounds subjective — how much do you want a coffee? — into a number that can be added, compared and drawn. And it does so without anybody having to justify their answer. Two people can hold wildly different WTPs for the same cup and both be right, because WTP is a fact about the person, not about the coffee.

The test of the definition. At exactly your WTP you are indifferent between buying and not. That indifference is what pins the number down: it is the price at which you stop, not the price you would like to pay. The two are very different, and the second is not a well-defined quantity at all.

Why everything else waits on this. Consumer surplus, total surplus, efficiency and dead-weight loss are all built out of WTP. If value could not be given a number, none of them could be computed.

Four regulars, one cart

Four regulars, one cart

A campus coffee cart. Four regulars, one cup each, tops:

Buyer	WTP for a cup
Maya	\$6.00
Omar	\$5.00
Priya	\$3.50
Ben	\$2.00

At $P = \$4.00$: Maya buys ($6.00 \geq 4.00$), Omar buys ($5.00 \geq 4.00$), Priya and Ben pass. $Q^d = 2$.

Price up, count down: that is a demand curve being born.

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Slide 7

Working through it

The setup. Four buyers, one cup each at most, with willingness to pay of \$6.00, \$5.00, \$3.50 and \$2.00.

The buying rule, and it is mechanical. Buy if $WTP \geq P$. Nothing psychological is going on: each buyer compares one private number with one public one.

Applying it at \$4.00. Maya ($6.00 \geq 4.00$) buys. Omar ($5.00 \geq 4.00$) buys. Priya ($3.50 < 4.00$) does not. Ben ($2.00 < 4.00$) does not. So $Q^d = 2$.

Where the law of demand comes from. Raise the price to \$5.50 and only Maya's number still covers it, so $Q^d = 1$. Drop it to \$3.00 and Priya joins, so $Q^d = 3$. Quantity demanded falls as price rises — and notice that *nothing was assumed* to make that happen. It fell out of four fixed private numbers being compared against a moving public one. The law of demand is a consequence, not a postulate.

The demand schedule, derived

The demand schedule, derived		
Buy if $WTP \geq P$. Sweep the price from high to low:		
Price range	Who buys	Q^d
above \$6.00	nobody	0
\$5.01 – \$6.00	Maya	1
\$3.51 – \$5.00	Maya, Omar	2
\$2.01 – \$3.50	Maya, Omar, Priya	3
\$2.00 and below	all four	4

Each price cutoff is somebody's WTP.

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Slide 8

Working through it

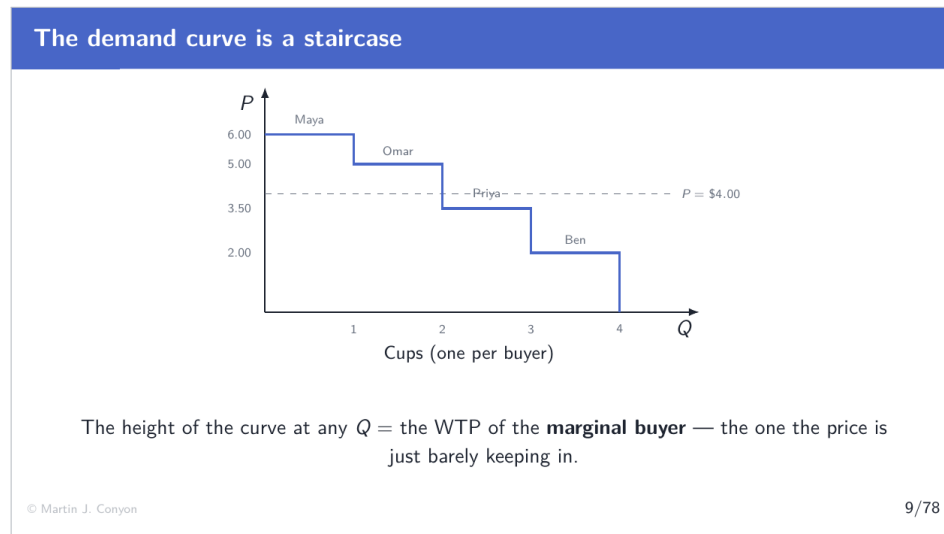
The method. Sweep the price from above everybody's WTP down to below everybody's, and at each level count who is still in.

Reading the rows. Above \$6.00 nobody's number covers the price, so $Q^d = 0$. Between \$5.01 and \$6.00 only Maya's does. Between \$3.51 and \$5.00, Maya and Omar. Between \$2.01 and \$3.50, add Priya. At \$2.00 and below, all four.

The observation that matters. **Every cutoff in that table is somebody's willingness to pay.** The schedule changes value at \$6.00, \$5.00, \$3.50 and \$2.00 — which is precisely the list of the four private numbers.

Why that is worth stopping on. It means a demand schedule is not an independent economic object that happens to slope downward. It is the buyers' private valuations, sorted from highest to lowest and read off in order. Anyone who knows the sorted list of WTPs knows the demand curve, and vice versa: the two contain exactly the same information.

The demand curve is a staircase



Slide 9

Working through it

Reading the figure. Quantity horizontally, price vertically. The blue staircase is the demand schedule from the previous frame, plotted. Each tread is one buyer, and the buyers appear tallest first because the schedule sorted them that way.

The dashed line. A price of \$4.00. It cuts the staircase after two treads, which is the $Q^d = 2$ computed earlier — the picture and the table agree by construction.

The reading to build into a habit. At any quantity, **the height of the demand curve is the willingness to pay of the marginal buyer** — the last one in, the one who would leave first if the price rose. At $Q = 2$ that height is \$5.00, which is Omar's number.

Why this vertical reading is the important one. Most of what follows measures *areas* under the demand curve, and an area is only meaningful once you know what the height means. Height is marginal value. Every triangle computed later is a sum of heights.

One buyer, several cups

One buyer, several cups

Why does *one person's* demand also slope down? Maya's morning, cup by cup:

Cup	Worth to Maya	Buys at \$4.00?
First	\$6.00	yes
Second	\$4.50	yes
Third	\$3.00	no
Fourth	\$1.50	no

Diminishing marginal utility. Each additional unit adds less satisfaction than the one before, so the dollar value a buyer places on successive units falls.

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Slide 10

Working through it

A gap in the story so far. Each buyer has been allowed one cup. That made the market demand curve slope down for a reason that had nothing to do with any individual: different people have different numbers. But an individual's own demand curve slopes down too, and this frame supplies that reason separately.

Maya's morning. Her first cup is worth \$6.00. The second, pleasant but less needed, \$4.50. The third, \$3.00. The fourth, \$1.50.

The principle. **Diminishing marginal utility:** each additional unit adds less satisfaction than the one before, so the dollar value placed on successive units falls.

Applying the same buying rule. At \$4.00 Maya buys cup one ($6.00 \geq 4$) and cup two ($4.50 \geq 4$), and stops — cup three at \$3.00 does not cover the price. She buys 2.

The parallel worth noticing. Maya's personal demand curve is a staircase descending through \$6.00, \$4.50, \$3.00, \$1.50 — structurally identical to the market staircase. In one case successive *buyers* are worth less; in the other, successive *units* are. The geometry does not care which.

Two reasons every demand curve falls

Two reasons every demand curve falls

As the price drops, quantity demanded rises through two doors:

Across buyers — lower prices recruit people whose WTP was below the old price.

Within each buyer — lower prices justify the next, less-valued unit.

Thousands of buyers $\times$ several units each $\Rightarrow$ the steps shrink until the staircase looks like a smooth line.

A demand curve is every unit anyone wants, ranked best-first.

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Slide 11

Working through it

Putting the two mechanisms together. A fall in price raises quantity demanded through two independent doors, and a real market uses both at once.

Across buyers. A lower price falls below the WTP of people who were previously excluded, and they enter. This is the mechanism of the four-regulars staircase.

Within each buyer. A lower price falls below the value an existing buyer places on their *next* unit, so they buy one more. This is the mechanism of Maya's morning.

Why real demand curves look smooth. With four buyers, the steps are large enough to see. With thousands of buyers, each weighing several units, there are tens of thousands of steps and each is tiny. Zoomed out, the staircase is indistinguishable from a straight line — which is why every diagram in this material draws one.

The sentence to keep. A demand curve is every unit anyone wants, ranked best-first. Nothing is lost in the smoothing: the line still records, at each quantity, the value of the marginal unit.

2. Part 2: From One Buyer to the Market

Write the equations

Write the equations

Two customers, personal weekly demands:

$$q_{\text{Noor}} = 6 - P \qquad q_{\text{Sam}} = 8 - 2P$$

(cups per week; each is zero once P passes the buyer's cutoff: \$6 for Noor, \$4 for Sam).

Read the coefficients:

- Intercepts 6 and 8: cups per week if coffee were free.
- Slopes 1 and 2: cups given up per dollar of price. **Sam is twice as price-sensitive.**

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Working through it

What this part is doing. Part 1 built a market demand curve by counting buyers. This part does it properly, with equations, so that market demand can be derived rather than tabulated.

The two personal demands.

$$q_{\text{Noor}} = 6 - P \qquad q_{\text{Sam}} = 8 - 2P$$

Reading every coefficient. The intercepts, 6 and 8, are cups per week at a price of zero — what each would drink if coffee were free. The slopes, 1 and 2, are cups given up per dollar of price. Sam's 2 against Noor's 1 says Sam is *twice as price-sensitive*.

The cutoffs, which matter more than they look. Each equation is only valid while it returns a non-negative quantity. Noor's hits zero at $P = 6$; Sam's at $P = 4$. Above those prices the buyer is simply out of the market, and their demand is zero, not negative. Sam leaving at \$4 while Noor stays is what puts a kink in the market curve two frames from now.

Plug the numbers in

Plug the numbers in			
P	Noor: $6 - P$	Sam: $8 - 2P$	Market Q^d
\$6	$6 - 6 = 0$	0 (out)	0
\$5	$6 - 5 = 1$	0 (out)	1
\$4	$6 - 4 = 2$	$8 - 8 = 0$	2
\$3	$6 - 3 = 3$	$8 - 6 = 2$	5
\$2	$6 - 2 = 4$	$8 - 4 = 4$	8
\$1	$6 - 1 = 5$	$8 - 2 = 6$	11

At each price, the market column is the *sum across the row*.

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Slide 16

Working through it

The procedure. Fix a price. Ask each buyer what they take at that price. Add the answers. Repeat.

Building the \$3 row. Noor: $6 - 3 = 3$ cups. Sam: $8 - 2(3) = 8 - 6 = 2$ cups. Market: $3 + 2 = 5$.

Building the \$5 row — the one with a trap in it. Noor: $6 - 5 = 1$. Sam: the formula would give $8 - 10 = -2$, which is meaningless. Sam is above his \$4 cutoff, so he buys **zero**. Market: $1 + 0 = 1$. Blindly substituting into a demand equation outside its valid range is the standard way to get this wrong.

Reading down the market column. 0, 1, 2, 5, 8, 11. Two things are visible. It rises as price falls, as it must. And the increments are uneven: 1, 1, 3, 3, 3. The jump from 1 to 3 happens exactly where Sam enters, and that unevenness is the kink about to appear on the graph.

Add quantities, never prices

Add quantities, never prices

Market demand. At each price, the sum of the quantities all individual buyers demand at that price. Fix P , add the q 's — sideways, never up.

Two price ranges, because Sam quits at \$4:

$$P > 4: Q^d = 6 - P \quad P \leq 4: Q^d = (6 - P) + (8 - 2P) = 14 - 3P$$

Check both ways at $P = 2$: table row $4 + 4 = 8$; equation $14 - 3(2) = 8$. They agree — if they did not, something upstream is wrong.

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Slide 17

Working through it

The rule, and its direction. Market demand is the sum of individual quantities *at a fixed price*. You add horizontally — sideways along the quantity axis — never vertically.

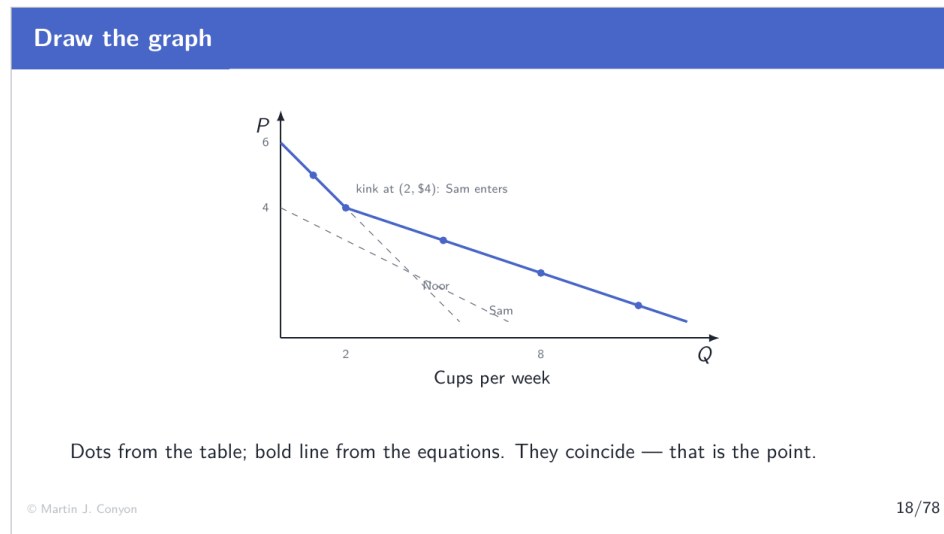
Why adding prices would be wrong. Prices are not shared out between buyers; every buyer faces the same price. What differs across buyers is how much they take at it. Adding two WTPs at a fixed quantity would answer a question nobody asked. (There is one context where curves are added vertically — public goods, where everybody consumes the same unit — and confusing the two is a standard error.)

Why there are two ranges. Above \$4, Sam is out and the market is Noor alone: $Q^d = 6 - P$. At or below \$4, both are in:

$$Q^d = (6 - P) + (8 - 2P) = 14 - 3P$$

The check. At $P = 2$ the table said $4 + 4 = 8$; the formula gives $14 - 3(2) = 8$. Two independent routes, one answer. If they had disagreed, exactly one of them contains an error and the disagreement tells you to go find it — which is worth more than a single unchecked number.

Draw the graph



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Working through it

What is drawn. The two thin lines are the personal demand curves. The bold line is their horizontal sum, which is market demand.

The upper segment. Above \$4 only Noor is in the market, so the bold line lies exactly on top of Noor's curve. Market demand *is* one person's demand up there.

The kink at (2, \$4). At \$4 Sam enters. Below that price every dollar of price cut releases Noor's 1 cup *and* Sam's 2, so the market curve suddenly gains 3 cups per dollar instead of 1. A curve that gains quantity faster is flatter, so the bold line bends outward at exactly Sam's cutoff.

The general lesson. Every kink in a market demand curve is a buyer entering or leaving. With two buyers you can see them; with thousands the kinks are everywhere and invisible, which is why aggregate curves are drawn straight.

The check, again. The dots are the rows of the table; the line is drawn from the equations. They coincide, because they are two portraits of one object.

Scaling up: our market

Scaling up: our market

A campus of thousands of Noors and Sams, summed sideways, gives the market we use all evening:

$$Q^d = 70 - 10P \qquad Q^s = 10P - 10$$

(Q in **thousands** of cups per week.)

Equilibrium:

$$70 - 10P = 10P - 10 \Rightarrow 20P = 80 \Rightarrow P^* = \$4, \quad Q^* = 70 - 10(4) = 30$$

Every number here comes out of these two equations.

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Slide 19

Working through it

The move. Everything so far has used two buyers so the arithmetic stayed visible. Now sum thousands of personal staircases across a whole campus. The kinks blur, and what remains is a straight line.

The market. With Q in **thousands** of cups per week:

$$Q^d = 70 - 10P \qquad Q^s = 10P - 10$$

Solving the equilibrium.

$$70 - 10P = 10P - 10 \implies 80 = 20P \implies P^* = 4$$

and $Q^* = 70 - 10(4) = 30$. Check in supply: $10(4) - 10 = 30$. ✓

Two numbers worth extracting now, because every area later uses them. Setting $Q^d = 0$ gives the *choke price* of \$7 — the height of the demand curve at its very first cup, and so the value of the most-wanted cup on campus. Setting $Q^s = 0$ gives \$1 — the cost of the cheapest cup anyone can supply.

What to carry forward. This demand curve is not an abstraction. It is thousands of personal

valuations, stacked tallest to shortest, and its height at any quantity is still the marginal buyer's WTP.

3. Part 3: Consumer Surplus

The buyer's gain has a name

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Consumer surplus (CS). What a buyer would have paid minus what the buyer actually pays: $CS = WTP - P$. The buyer's gain from the trade, in dollars.

The four regulars at $P = \$3.50$:

Buyer	WTP	Buys?	CS = WTP - 3.50
Maya	\$6.00	yes	$6.00 - 3.50 = \$2.50$
Omar	\$5.00	yes	$5.00 - 3.50 = \$1.50$
Priya	\$3.50	yes	$3.50 - 3.50 = \$0$
Ben	\$2.00	no	—
Total CS			\$4.00

The **marginal buyer** (Priya) gains exactly zero.

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Working through it

The definition. **Consumer surplus** is what a buyer would have paid minus what they actually pay: $CS = WTP - P$. It is the buyer's gain from the trade, in dollars.

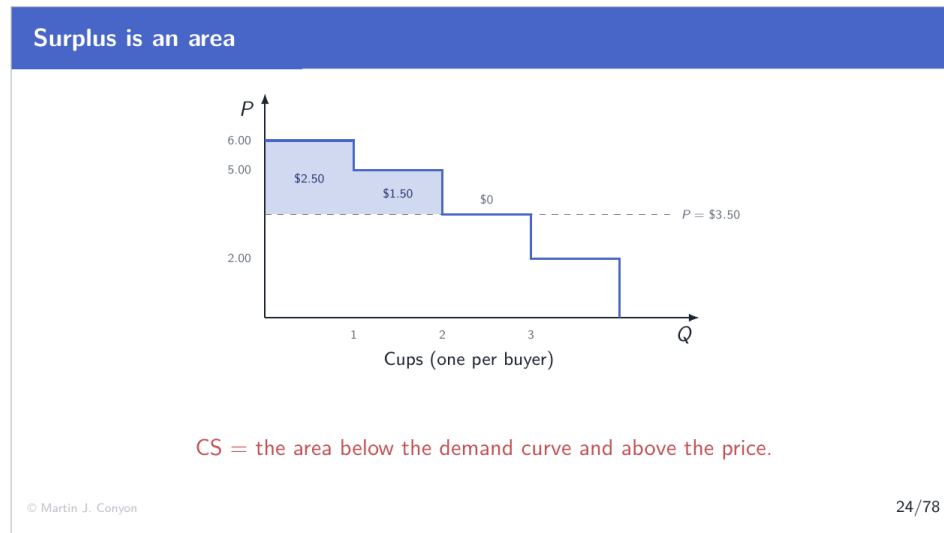
Why a voluntary trade must produce it. Nobody buys at a price above their WTP. So every completed purchase has $WTP \geq P$, and the surplus is never negative. Trade does not merely move money about; it leaves the buyer strictly better off unless they were exactly at the margin.

The table at $P = \$3.50$, row by row. Maya: $6.00 - 3.50 = \$2.50$. Omar: $5.00 - 3.50 = \$1.50$. Priya: $3.50 - 3.50 = \$0$. Ben does not buy at all, so he has no surplus — not a negative one. Total: $2.50 + 1.50 + 0 = \$4.00$.

The marginal buyer earns exactly nothing. Priya's WTP equals the price, so she is precisely indifferent between buying and not. This is not a quirk of the numbers; it is what *being* the

marginal buyer means, and the same fact reappears for the marginal seller in Part 4.

Surplus is an area



Slide 24

Working through it

From a column of numbers to a shape. Each buyer's surplus is the vertical gap between their step on the staircase and the price line. Maya's block is tall (\$2.50), Omar's shorter (\$1.50), Priya's has no height at all.

Why the blocks are areas rather than lengths. Each buyer takes one cup, so each block is one unit wide. Height $\times$ width = surplus, and the widths happen to be 1. When buyers take several units the widths grow, and the area interpretation is the one that survives.

The general statement.

$$\text{CS} = \text{area below the demand curve and above the price,}$$

out to the quantity actually traded.

Why that sentence is worth memorizing exactly. It is stated in terms of the curve and the price alone, so it works whatever shape the demand curve has and however many buyers there are. When the staircase smooths into a straight line the blocks become a triangle, and nothing about the definition changes — only the formula for computing the area.

The whole market's CS

The whole market's CS

Our market at $P^* = \$4$: height of demand runs from \$7 (the first, most-valued cup) down to \$4 (the marginal cup at $Q = 30$).

Triangle: base = 30, height = $7 - 4 = 3$.

$$CS = \frac{1}{2} \times 30 \times (7 - 4) = \frac{1}{2} \times 30 \times 3 = 45$$

\$45,000 per week of buyer gain — created by the market's mere existence.

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Working through it

Applying the area rule to the campus market. Consumer surplus is the region below demand and above the price, from $Q = 0$ out to $Q^* = 30$. With a straight demand curve that region is a triangle.

Its two dimensions. The base is the quantity traded, 30 thousand cups. The height is the vertical distance from the price to the top of the demand curve: the demand curve reaches \$7 at $Q = 0$ (the choke price), and the price is \$4, so the height is $7 - 4 = 3$ dollars.

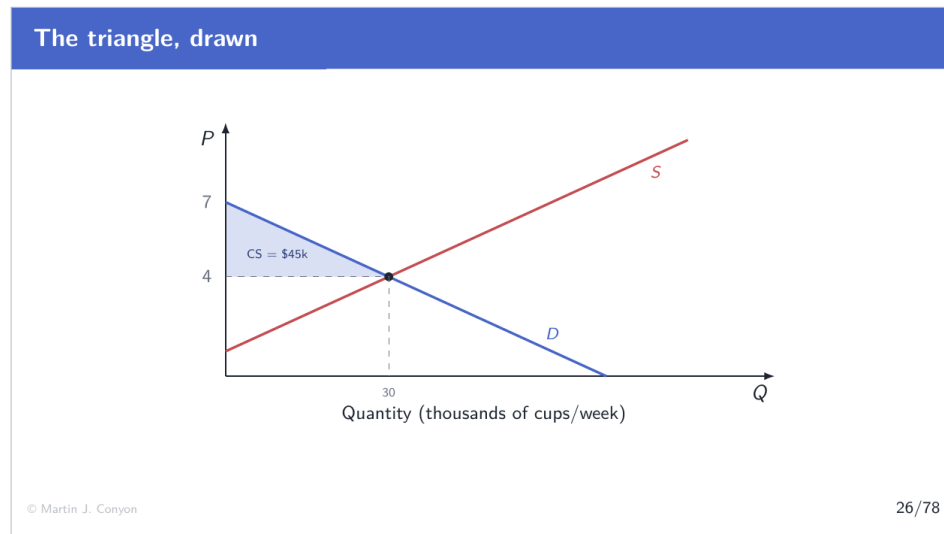
The area.

$$CS = \frac{1}{2} \times 30 \times 3 = 45$$

Units, which are easy to lose. Q is in thousands of cups and the height is in dollars per cup, so the product is in thousands of dollars: **\$45,000 per week**.

What the number is not. It is not revenue (that is $4 \times 30 = \$120k$) and it is not spending. It is the value buyers receive *over and above* everything they hand to sellers — the part of the transaction that does not show up in anybody's accounts.

The triangle, drawn



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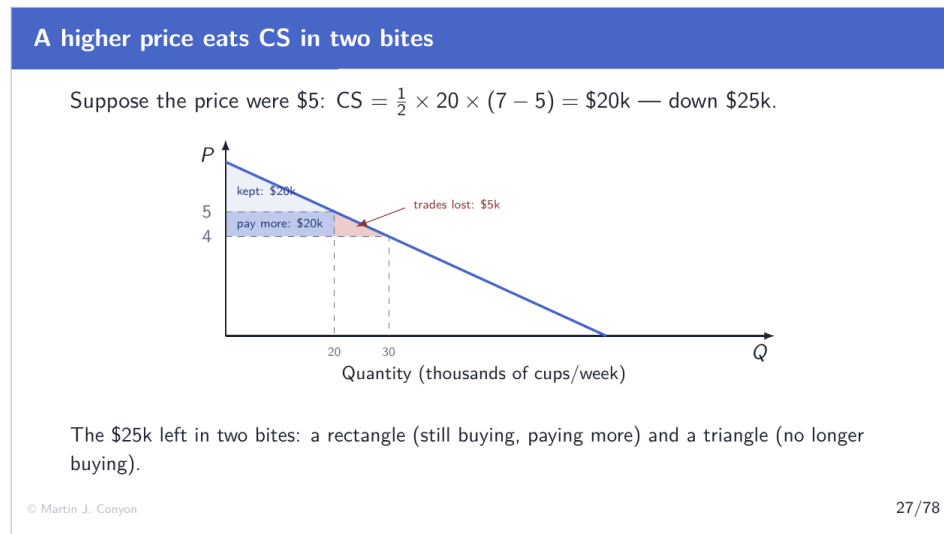
Working through it

Locating the shape. The blue triangle is bounded by three things: the demand curve above, the \$4 price line below, and the vertical axis on the left. It runs out to $Q^* = 30$, where demand meets the price and the triangle closes to a point.

Why it is tall on the left and pinched on the right. The left edge represents the earliest, most-wanted cups, whose buyers valued them near \$7 and paid \$4 — roughly \$3 of surplus each. Moving right, each successive cup is wanted slightly less, so the gap narrows. At the marginal cup the gap is zero, which is the marginal buyer earning nothing.

The habit to build. From here on, whenever a demand curve and a price appear together, complete this triangle in your head before reading anything else off the diagram. Almost every welfare question that follows is answered by comparing two versions of it.

A higher price eats CS in two bites



Slide 27

Working through it

The new triangle. At $P = \$5$, quantity falls to $Q^d = 70 - 10(5) = 20$, and

$$CS = \frac{1}{2} \times 20 \times (7 - 5) = \$20k,$$

down \$25k from \$45k.

Where the \$25k went — and it left by two different doors.

Bite one, the rectangle: \$20k. The 20 thousand cups still bought are each now paying \$1 more. $20,000 \times \$1 = \$20k$. These buyers did not leave; they simply keep less of the gain. Note that this money is not destroyed: the sellers receive it. It is a *transfer*.

Bite two, the triangle: \$5k. Ten thousand cups are no longer bought at all — those valued between \$4 and \$5. The surplus on them ran from \$1 down to \$0, so the lost area is $\frac{1}{2} \times 10 \times 1 = \$5k$. And this money goes nowhere. Nobody receives it, because the trades do not happen.

Why the distinction is the whole session in miniature. A transfer moves surplus between people; a lost trade destroys it. Everything from Part 5 onward turns on telling the two apart.

What CS is actually measuring

What CS is actually measuring

If this market vanished tomorrow, its buyers would be exactly **\$45,000 a week poorer** — that is what CS prices.

It is value *as the buyers themselves judge it*: their own WTP, not anyone's opinion of what coffee should be worth.

Next: the sellers' side of the same handshake.

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Slide 28

Working through it

The question consumer surplus answers. Not “how much do buyers spend” and not “how happy are buyers” — but precisely: *if this market vanished tomorrow, by how many dollars would its buyers be worse off?* For the campus market the answer is \$45,000 a week.

Why that framing is useful. It converts a welfare judgment into an arithmetic one, and it makes different policies comparable. A rule that costs buyers \$10k of surplus is worse for them than one costing \$3k, and the comparison needs no opinion about coffee.

Whose valuation is being used. The buyers' own. CS is assembled entirely out of willingness to pay, so it never substitutes anybody's view of what coffee *ought* to be worth for what buyers say it is worth to them.

The limitation, stated honestly. That same feature means WTP reflects ability to pay as well as intensity of desire. A wealthy buyer's \$6 and a poor buyer's \$6 count identically in this arithmetic, though they may represent very different sacrifices. Consumer surplus measures efficiency, not fairness, and Part 5 states that boundary explicitly.

4. Part 4: Producer Surplus

The seller's walk-away price

The seller's walk-away price

Cost. Everything a seller gives up to supply one more unit — materials, time, and the next-best use of both (opportunity cost). Cost is the seller's walk-away price: the lowest price at which the unit is worth supplying.

Three carts, cost per cup:

Cart	Cost per cup	Sells at $P = \$3.00$?
Rosa	\$1.50	yes
Dev	\$2.50	yes
Lena	\$3.50	no

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Working through it

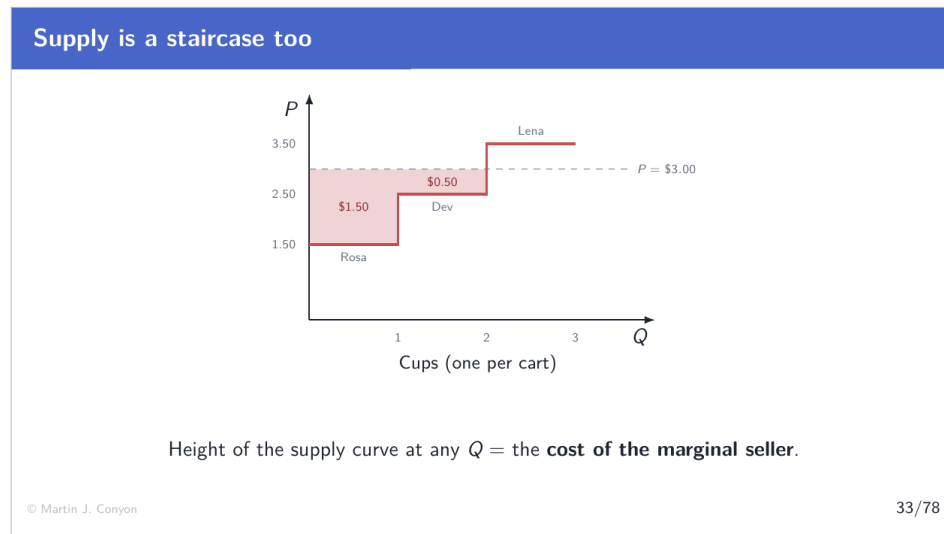
The mirror of willingness to pay. A seller's **cost** for a unit is everything supplying it uses up — and *everything* is meant literally.

What goes into it. Materials (beans, milk, a cup), the seller's time, and the value of the next-best use of both. That last item is opportunity cost, and leaving it out is the standard error: an owner who works the cart for an hour gives up whatever that hour could have earned elsewhere, whether or not any money changes hands.

Why cost is a walk-away price. At a price below cost, supplying the unit makes the seller worse off than not supplying it. So cost is the minimum price at which the unit is worth producing — exactly as WTP is the maximum at which it is worth buying.

The three carts at $P = \$3.00$. Rosa's cup costs \$1.50, so she supplies. Dev's costs \$2.50, so he supplies. Lena's costs \$3.50, so she does not — and that is the *correct* decision, not a missed opportunity. Selling at \$3.00 a cup that consumes \$3.50 of resources destroys fifty cents of value.

Supply is a staircase too



Slide 33

Working through it

Sorting the other way. Buyers were sorted highest WTP first, because the highest-value buyers are served first. Sellers are sorted *cheapest first*, because the lowest-cost sellers supply first. The result is a rising staircase — the exact mirror of the falling one.

Reading the height. At any quantity, **the height of the supply curve is the cost of the marginal seller** — the last one the price is persuading. That is the counterpart of the vertical reading built in Part 1, and it is what makes areas under the supply curve meaningful.

At $P = \$3.00$. The price line cuts the staircase after two treads, so $Q^s = 2$. Rosa clears $3.00 - 1.50 = \$1.50$ on her cup; Dev clears $3.00 - 2.50 = \$0.50$; Lena, whose tread is above the line, supplies nothing and clears nothing.

The shaded blocks. They sit *between the price line and the staircase*, which is the same construction as the buyers' blocks, reflected. For buyers the surplus hung below the curve and above the price; here it hangs above the curve and below the price.

Producer surplus, defined and scaled

Producer surplus, defined and scaled

Producer surplus (PS). What a seller is paid minus the seller's cost: $PS = P - \text{cost}$. The area below the price and above the supply curve.

Our market at $P^* = \$4$: supply height runs \$1 (cheapest cup) up to \$4 (marginal cup at $Q = 30$).

$$PS = \frac{1}{2} \times 30 \times (4 - 1) = \frac{1}{2} \times 30 \times 3 = 45$$

\$45,000 per week of seller gain. The marginal seller — like the marginal buyer — gains zero.

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Working through it

The definition, obtained by symmetry. **Producer surplus** is what a seller is paid minus what the unit cost them: $PS = P - \text{cost}$, summed over units sold. Geometrically: the area below the price and above the supply curve.

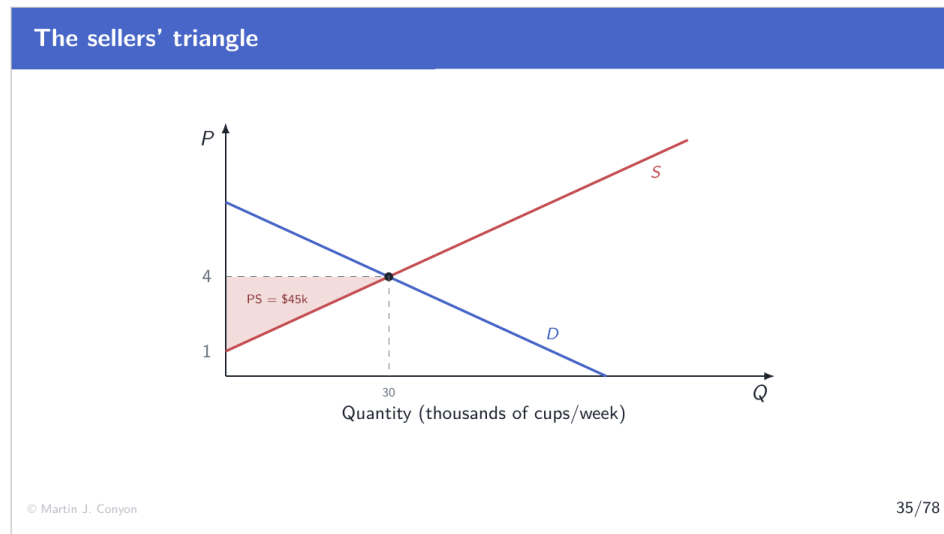
Applying it to the campus market. The region is a triangle with base $Q^* = 30$ and height equal to the distance from the supply curve's starting height to the price. Setting $Q^s = 10P - 10 = 0$ gives $P = 1$, so the cheapest cup costs \$1 to supply, and the height is $4 - 1 = 3$.

$$PS = \frac{1}{2} \times 30 \times 3 = 45 \quad \implies \quad \$45,000 \text{ per week}$$

The marginal seller again. At $Q = 30$ the supply curve's height is exactly \$4, the price. The last seller's cost equals what they receive, so they gain nothing and are indifferent about supplying at all.

Why both margins are worth zero. At the equilibrium quantity the marginal buyer's value and the marginal seller's cost are both equal to the price — and therefore to each other. That coincidence is the reason the market stops exactly there, and Part 5 makes it the center of the efficiency argument.

The sellers' triangle



Slide 35

Working through it

Locating the shape. The red triangle is bounded by the supply curve below, the \$4 price line above, and the vertical axis on the left, running out to $Q^* = 30$.

Why it is tall on the left. The left edge is the cheapest cups, costing about \$1 to produce and selling for \$4 — roughly \$3 of surplus each. Moving right, each successive cup costs more to supply, so the margin narrows, closing to zero at the marginal cup.

The two triangles together. The blue region above the price and the red region below it now cover the whole area between the demand curve and the supply curve, out to the equilibrium quantity. Nothing is left over and nothing is double-counted, and that is not a coincidence — the price line is simply the boundary that divides one wedge into two.

What the price line does and does not do. Moving it up or down redistributes area between the two triangles. It does not, at the equilibrium quantity, change their total. Part 5 turns that observation into the main result.

Two triangles, one symmetry

Two triangles, one symmetry		
	Consumer surplus	Producer surplus
Private number	WTP (max they'd pay)	cost (min they'd take)
Gain per unit	$WTP - P$	$P - \text{cost}$
On the diagram	below D , above price	above S , below price
At the margin	zero (marginal buyer)	zero (marginal seller)
Our market	\$45k/week	\$45k/week

Equal here by construction of the example — not a law. The symmetry of *logic* is the law.

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Slide 36

Working through it

Reading the table as a mirror. Every row is the same idea reflected in the price line.

- **The private number.** Buyers hold a maximum (WTP); sellers hold a minimum (cost). Both are facts about the participant, not the good.
- **The gain per unit.** $WTP - P$ against $P - \text{cost}$. Each is the distance between the private number and the price, measured in the direction that makes it positive.
- **The area.** Below D and above the price; above S and below the price. The same region, on opposite sides of the same line.
- **At the margin.** Zero for both, because at the margin the private number equals the price.

The warning in the last row. Both surpluses come to \$45k here, and that is an artifact of the example: the demand and supply curves were chosen with matching slopes. **The equality is not a law.** Make demand steeper and buyers capture more; make supply steeper and sellers do. What *is* a law is the structure of the argument, which is identical on both sides.

5. Part 5: Is the Market Any Good?

Add the gains: total surplus

Add the gains: total surplus

Total surplus (TS). Consumer surplus plus producer surplus: the entire gain the market creates for its participants.

$$TS = \underbrace{(\text{value to buyers} - \text{paid})}_{CS} + \underbrace{(\text{received} - \text{cost})}_{PS} = \text{value to buyers} - \text{cost to sellers}$$

The money paid **cancels** — it changed hands, it didn't vanish.

Our market: $TS = 45 + 45 = \$90,000$ per week.

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Working through it

The definition. **Total surplus** is consumer surplus plus producer surplus — the entire gain the market creates for its participants.

The cancellation, which is the point of the frame. Write the two out in full:

$$CS = (\text{value to buyers}) - (\text{paid}) \quad PS = (\text{received}) - (\text{cost to sellers})$$

What buyers pay is what sellers receive. It appears with a minus sign in one and a plus sign in the other, so when the two are added it disappears:

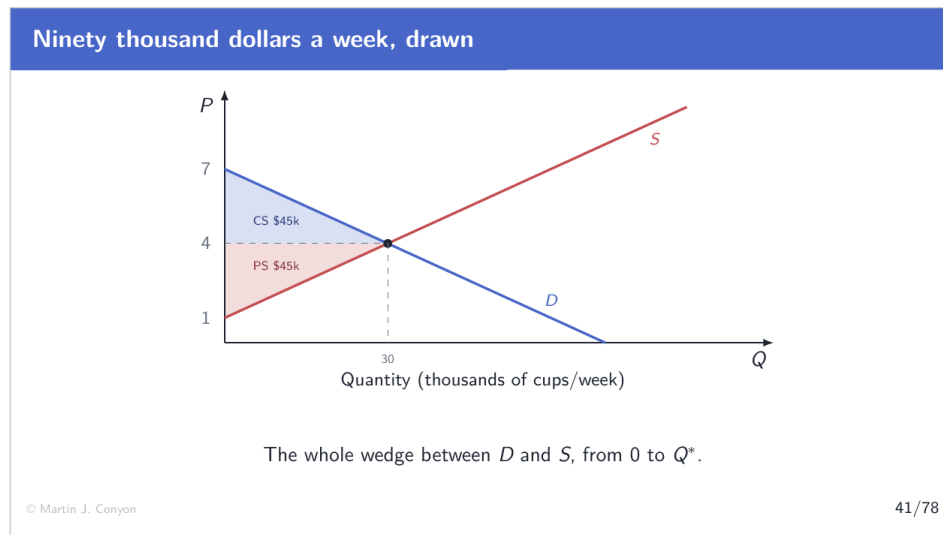
$$TS = (\text{value to buyers}) - (\text{cost to sellers})$$

Why that matters more than it looks. Money changing hands is a *transfer*: it moves surplus between participants without creating or destroying any. What total surplus measures is the genuinely real thing — the satisfaction obtained set against the resources used up.

The number. $45 + 45 = \$90,000$ per week.

And the question the number invites. Could any regulator, planner or committee rearrange this market — who buys, who sells, how much trades — and get more than 90? The next three frames answer no, three times over.

Ninety thousand dollars a week, drawn



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Working through it

One shape, not two. The blue and red triangles together fill the whole wedge between the demand curve and the supply curve, from $Q = 0$ out to $Q^* = 30$. That wedge is total surplus.

Why it is a wedge. At each quantity, the height of demand is what the marginal cup is worth to a buyer, and the height of supply is what it costs to make. The vertical gap between them is therefore the value created by that cup. Summing the gap across all cups traded gives the area between the curves — and the wedge is the picture of that sum.

What the price line does. It divides the wedge into the two triangles. Nothing more. Slide it up and buyers keep less while sellers keep more; the total is unchanged so long as the quantity traded is unchanged.

Where the wedge closes. At $Q^* = 30$ the two curves meet, so the gap is zero: the last cup traded is worth exactly what it costs. That is the marginal buyer and the marginal seller

both earning nothing, seen as geometry.

What to watch for later. Every failure in the second half of the session shows up as a piece missing from this wedge.

The right buyers, the right sellers

The right buyers, the right sellers

At $P^* = \$4$, with no one directing anything:

- Every buyer with $WTP \geq \$4$ buys — the 30,000 **highest-value** cups are the ones consumed.
- Every seller with $\text{cost} \leq \$4$ sells — the 30,000 **cheapest** cups are the ones produced.

Reassign any cup from a buyer to a non-buyer, or a seller to a non-seller, and total surplus **falls**.

The price is a sorting machine. Nobody built it; nobody runs it.

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Working through it

The first of three efficiency claims: the market allocates to the right people.

On the buying side. A price of \$4 admits exactly those buyers whose WTP is at least \$4, and excludes the rest. Because the demand curve sorts buyers highest-value first, the 30 thousand cups consumed are the 30 thousand *most valued* cups on campus.

On the selling side. The same price admits exactly those sellers whose cost is at most \$4. Because supply sorts cheapest first, the cups produced are the 30 thousand *cheapest to produce*.

Testing it by trying to do better. Take a cup from someone who bought it ($WTP \geq \$4$) and give it to someone who did not ($WTP < \$4$). Value has moved from a higher use to a lower one, so total surplus falls. Now make a non-producing cart supply a cup in place of a producing one: its cost exceeds \$4 while the displaced cart's was below, so resources used rise and surplus falls again. **Every reshuffle loses money.**

And nobody organized it. No official collected WTPs, ranked them, or issued permits. One number did the sorting, and it emerged from the market itself.

The right amount

The right amount			
Value of the marginal cup: $(70 - Q)/10$. Its cost: $(Q + 10)/10$.			
Q (000s)	Marginal value	Marginal cost	Gap
20	$\frac{70-20}{10} = 5.00$	$\frac{20+10}{10} = 3.00$	+2.00
25	$\frac{70-25}{10} = 4.50$	$\frac{25+10}{10} = 3.50$	+1.00
30	$\frac{70-30}{10} = 4.00$	$\frac{30+10}{10} = 4.00$	0.00
35	$\frac{70-35}{10} = 3.50$	$\frac{35+10}{10} = 4.50$	-1.00
40	$\frac{70-40}{10} = 3.00$	$\frac{40+10}{10} = 5.00$	-2.00

Below 30: stopping early **forgoes** gains. Past 30: each extra cup **destroys** value.

TS peaks at $Q = 30$ — the market's own unsupervised choice.

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Working through it

The second claim: the market trades the right quantity — and the table demonstrates it arithmetically, with no calculus anywhere.

Where the two formulas come from. Invert each curve. From $Q^d = 70 - 10P$, the height of demand at quantity Q is $P = \frac{70-Q}{10}$: the value of the marginal cup. From $Q^s = 10P - 10$, the height of supply is $P = \frac{Q+10}{10}$: the cost of the marginal cup.

Reading the rows. At $Q = 20$: value \$5.00, cost \$3.00, so the next cup creates \$2.00. Stopping at 20 leaves that on the table. At $Q = 25$ the gap has narrowed to \$1.00. At $Q = 30$ it is exactly zero.

Past 30 the sign flips. At $Q = 35$: value \$3.50, cost \$4.50. Making that cup consumes \$4.50 of resources to produce something worth \$3.50 — it *destroys* a dollar of value.

The conclusion. Total surplus rises with every cup up to 30 and falls with every cup after it, so it peaks exactly at 30 — which is where the unregulated market settled by itself. **Too little and too much are both costly**, and that evenhandedness is what makes the same

triangle work for externalities later.

Say it properly: efficiency

Say it properly: efficiency

Efficiency. An allocation is efficient when it maximizes total surplus: no change in who buys, who sells, or how much is traded can create more value.

What efficiency does **not** say: nothing about *who gets* the surplus.

Equality — how the gains are shared — is a separate question, and a *normative* one. Our instrument certifies the size of the pie, not the fairness of the slices.

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Working through it

The definition. An allocation is **efficient** when it maximizes total surplus: no change in who buys, who sells, or how much trades can create more value.

What has been established. The competitive equilibrium is efficient. The right buyers get the good, the right sellers produce it, and the quantity stops exactly where marginal value meets marginal cost.

What efficiency does not say — and this is the part most often overstated. It is completely silent about *who gets* the surplus. A market can be perfectly efficient while one side captures nearly all of the wedge. Efficiency is a statement about the *size* of the pie, not about how it is cut.

Why that boundary is worth policing. “Efficient” and “good” are different claims. The first is positive and this analysis certifies it; the second is normative and this analysis cannot reach it. Equality of the shares is a question about values, and answering it is a citizen’s job, not a diagram’s.

A practical consequence. A policy that shrinks total surplus may still be worth adopting if

it redistributes toward people you think deserve more — but the trade-off should be made with the cost visible, which is precisely what the surplus arithmetic supplies.

Adam Smith saw it in 1776

Adam Smith saw it in 1776

"It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest. ... [Each] is led by an invisible hand to promote an end which was no part of his intention."

— Adam Smith, *The Wealth of Nations* (1776)

In our language: **self-interest, filtered through a market price, maximizes total surplus.**

We just proved with a table what Smith could only name.

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Working through it

The quotation. Smith's butcher, brewer and baker serve their customers from self-interest rather than benevolence, and are "led by an invisible hand to promote an end which was no part of [their] intention."

What the metaphor was, and was not. Smith identified the phenomenon: a decentralized, self-interested system producing a coordinated outcome nobody intended. He did not have the apparatus to say precisely what outcome, or to prove it was the best available.

Translating it into the language of these slides. The hand is the price. It sorts buyers by willingness to pay, sorts sellers by cost, and halts trade exactly where marginal value equals marginal cost. The end promoted is the maximization of total surplus.

Why the table is the important part. Two frames ago you verified this numerically: gaps positive below $Q = 30$, zero at 30, negative above. That is a proof, not an image. Smith named something he could see happening; the surplus arithmetic is what turned the observation into a theorem.

The honest qualification. Smith's claim is often quoted as though markets always work. The next frame lists the two assumptions this proof quietly used, and the rest of the session takes one of them away.

The fine print

The fine print

That verdict leaned on two quiet assumptions:

1. **Competition.** No single seller or buyer sets the price. (When this fails — market power — is a story for later in the material.)
2. **Everyone affected is at the table.** The trade touches nobody but its own buyer and seller.

Market failure. A situation in which an unregulated market's equilibrium fails to maximize total surplus.

From here on: assumption 2 breaks.

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Working through it

Two assumptions were doing silent work.

1. Competition. No single buyer or seller is large enough to move the price. If one seller can set it, the quantity traded is no longer where marginal value meets marginal cost, and the efficiency result fails. That is market power, and it is taken up when we come to monopoly.

2. Everybody affected is at the table. Every dollar of the \$90,000 belonged to a buyer or a seller in this market. Nobody outside it was helped or harmed. This assumption is so natural that it is easy to miss being made at all.

The definition that follows. **Market failure:** a situation in which an unregulated market's equilibrium fails to maximize total surplus.

Why the second assumption is the interesting one. It fails constantly and quietly. A trade in firewood affects the people downwind. A bike ride affects the drivers left in the traffic. In

neither case is anything wrong with the buyer or the seller — they are behaving exactly as before. What has changed is that the market's books no longer record everyone the trade reaches, and the remaining parts show what that does to the quantity traded.

6. Part 6: Costs That Spill Over

The bystander

The bystander

Externality. A cost or benefit from a trade that lands on someone who is not a party to it — a bystander, neither paying nor paid.

Negative — harm spills out: ▪ wood smoke drifting downhill ▪ a gas leaf blower at 7 a.m. ▪ one more SUV at rush hour	Positive — benefit spills out: ▪ a rain garden soaking up street runoff ▪ a neighbor's well-kept facade ▪ every commuter who bikes instead
--	--

The market's books only ever record the two people shaking hands.

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Working through it

The definition. An **externality** is a cost or benefit from a trade that lands on somebody who is not a party to it — a bystander, neither paying nor paid, and never asked.

The test to apply, and it is the one students skip. Name the bystander. If you cannot point to a specific third party who is affected, there is no externality, however large the effects on the buyer and seller may be.

Negative: harm spills out. Wood smoke drifting downhill onto households that bought no wood. A gas leaf blower at 7 a.m. reaching sleeping neighbors who bought no landscaping. One more SUV in rush hour, slowing every other driver.

Positive: benefit spills out. A rain garden absorbing runoff that would have flooded the whole street. A well-kept facade supporting the neighbors' resale values. A commuter on a

bicycle, who is one car fewer in everyone else's way.

Why the market cannot see any of it. A price is agreed between two parties, and it reflects what those two parties care about. The bystander is not at the negotiation, has no way to bid, and appears nowhere in the accounts. The market is not making an error — it is faithfully solving a problem that leaves somebody out.

A mountain town that burns wood

A mountain town that burns wood

The market for firewood bundles in a valley town:

$$Q^d = 100 - 10P \qquad Q^s = 10P - 20$$

(Q in **hundreds** of bundles per week.)

Equilibrium:

$$100 - 10P = 10P - 20 \Rightarrow 20P = 120 \Rightarrow P^* = \$6, \quad Q^* = 100 - 10(6) = 40$$

By everything so far: efficient. **For buyers and sellers.**

But every bundle burned sends smoke down the valley.

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Working through it

The market. Firewood bundles in a valley town, with Q in **hundreds** of bundles per week:

$$Q^d = 100 - 10P \qquad Q^s = 10P - 20$$

Solving.

$$100 - 10P = 10P - 20 \Rightarrow 120 = 20P \Rightarrow P^* = 6$$

and $Q^* = 100 - 10(6) = 40$, that is 4,000 bundles a week. Check in supply: $10(6) - 20 = 40$.

✓

The verdict everything so far would deliver. This equilibrium maximizes the total surplus of the people trading firewood. Right buyers, right sellers, right quantity. If burning wood

affected nobody else, the analysis would stop here with another gold star.

The fact that changes the question. Every bundle burned produces smoke, and the smoke drifts down the valley onto households who bought nothing and sold nothing. Assumption 2 of the fine print has just failed, and the next frame prices the failure.

What a bundle truly costs

What a bundle truly costs

Suppose the smoke from each bundle does **\$2.00** of harm to bystanders (health, laundry, cleaning).

Social cost. Private cost plus external cost: what one more unit costs *society*, counting the bystanders.

Q	WTP	Private cost	Social cost	Society's gain
20	$\frac{100-20}{10} = 8.00$	$\frac{20+20}{10} = 4.00$	$4.00 + 2 = 6.00$	+2.00
30	$\frac{100-30}{10} = 7.00$	$\frac{30+20}{10} = 5.00$	$5.00 + 2 = 7.00$	0.00
40	$\frac{100-40}{10} = 6.00$	$\frac{40+20}{10} = 6.00$	$6.00 + 2 = 8.00$	-2.00

The market stops where *private* books balance ($Q = 40$). Society's books balance at $Q = 30$.

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Working through it

Putting a number on the smoke. Suppose each bundle burned does \$2.00 of harm to bystanders — medical costs, cleaning, laundry — spread across the town. The figure is an estimate; the method does not depend on its exact size.

The definition. **Social cost** is private cost plus external cost: what one more unit costs *society*, counting everybody.

Reading the table. The columns are the same marginal-value and marginal-cost heights as in Part 5, with one added.

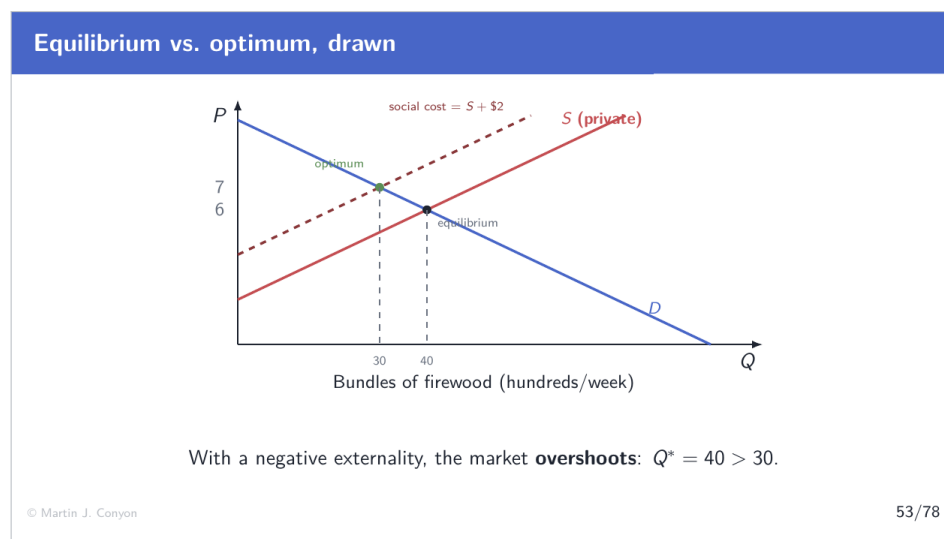
- At $Q = 20$: buyers value the bundle at \$8.00; private cost \$4.00; social cost $4.00 + 2 = \$6.00$. Society gains \$2.00, so this bundle should be burned.
- At $Q = 30$: value \$7.00, social cost \$7.00. The gain is zero. This is where a town

accountant counting everybody would stop.

- At $Q = 40$: value \$6.00, social cost \$8.00. Society *loses* \$2.00 on this bundle.

The two answers, and the gap between them. The market's books balance where value meets *private* cost: $Q = 40$. Society's books balance where value meets *social* cost: $Q = 30$. The market trades a thousand bundles too many, every week.

Equilibrium vs. optimum, drawn



Slide 53

Working through it

The third curve. The dashed line is social cost: the private supply curve lifted vertically by \$2 at every quantity. It is parallel to supply, because the external harm is the same \$2 on every bundle.

Two intersections, two answers.

- Demand meets *private* supply at the black dot: $Q^* = 40$, $P^* = \$6$. This is what the market does.
- Demand meets *social cost* at the green dot: $Q = 30$, at a height of \$7. This is what the market should do.

Solving for the optimum algebraically. Set marginal value equal to marginal social cost:

$$\frac{100 - Q}{10} = \frac{Q + 20}{10} + 2 \implies 100 - Q = Q + 40 \implies Q = 30$$

Which way the market errs, and why. It **overshoots**. The curve sellers actually respond to sits \$2 below the true cost curve, so trade continues past the point where the true curves cross. Nobody is behaving badly: the \$2 is real, and nobody at the till is asked to pay it.

The thousand bundles too many

The thousand bundles too many			
Bundles between the optimum and the equilibrium, one at a time:			
Bundle (00s)	Worth to buyer	Social cost	Society loses
32	$\frac{100-32}{10} = 6.80$	$\frac{32+40}{10} = 7.20$	0.40
36	$\frac{100-36}{10} = 6.40$	$\frac{36+40}{10} = 7.60$	1.20
40	$\frac{100-40}{10} = 6.00$	$\frac{40+40}{10} = 8.00$	2.00

Buyer and seller both said yes to every one of these — their private gain is real. The \$2 landing on the bystanders outweighs it.

Each of these trades *destroys* value — it just destroys it off the books.

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Working through it

Examining the overshoot bundle by bundle. These are the bundles between the optimum (3,000) and the equilibrium (4,000).

Bundle 3,200. Worth $\frac{100-32}{10} = \$6.80$ to its buyer. Social cost $\frac{32+40}{10} = \$7.20$. Burning it destroys \$0.40 of value.

Bundle 4,000. Worth \$6.00; social cost \$8.00. It destroys \$2.00.

Why these trades happen anyway — the crucial point. Look at the private arithmetic. At bundle 4,000 the buyer values it at \$6.00 and pays \$6.00; the seller's private cost is \$6.00 and receives \$6.00. *Both are content.* Neither is being foolish or dishonest, and no regulation of

their behavior toward each other would change anything.

Where the loss comes from. The arithmetic only turns negative when the bystanders' \$2 is added, and the bystanders are not in the room and have no way to enter it. The market is functioning perfectly as a machine for maximizing the welfare of the people it can see. The problem is the guest list.

Name the damage: deadweight loss

Name the damage: deadweight loss

Deadweight loss (DWL). Total surplus lost because the quantity traded is not the efficient quantity — value destroyed by trading too much or too little, collected by no one.

Here: losses on the extra bundles grow from \$0 (at 3,000) to \$2.00 (at 4,000) — a triangle:

$$DWL = \frac{1}{2} \times 10 \times \$2.00 = 10 \Rightarrow \text{\textcolor{red}{\$1,000 per week}}$$

Too much trade destroys value as surely as too little. The triangle measures either.

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Working through it

The definition, in its general form. **Deadweight loss** is total surplus lost because the quantity traded is not the efficient quantity — value destroyed by trading too much or too little, and collected by nobody.

The evenhandedness is deliberate. Nothing in the definition says “too much”. Part 5 showed that surplus falls on *both* sides of the efficient quantity, so a market that trades too little destroys value exactly as a market that trades too much does. Part 7 uses the other direction.

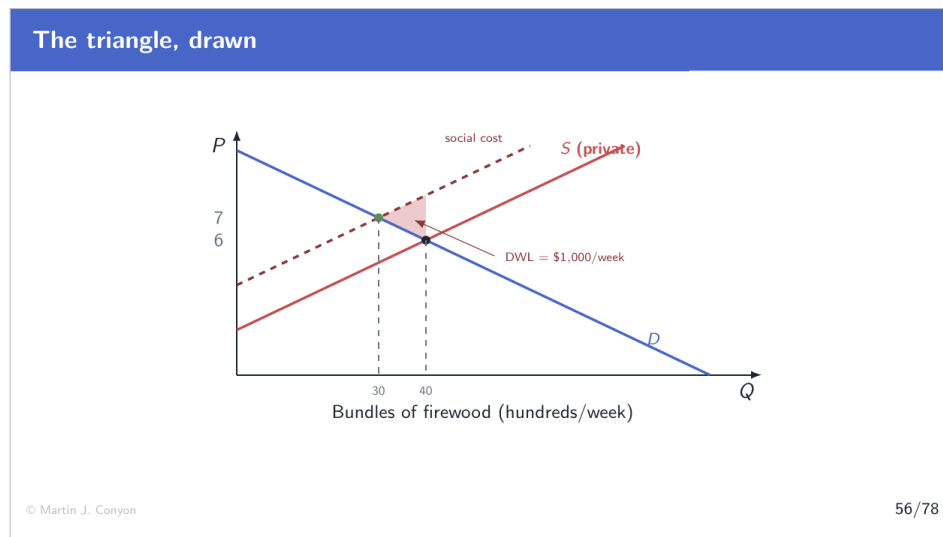
Computing it here. Across the excess bundles the per-bundle loss grows steadily from \$0 (at 3,000, where value still equalled social cost) to \$2.00 (at 4,000). A quantity rising linearly

from 0 to 2 across 1,000 bundles sums to a triangle:

$$DWL = \frac{1}{2} \times 1,000 \times \$2.00 = \text{\$1,000 per week}$$

Why “collected by nobody” is the operative phrase. This is not money moving from the town to the wood sellers. The buyer paid, the seller was paid, and the bystanders absorbed harm larger than the gain. The net is a loss with no corresponding gain anywhere — which is what distinguishes it from a transfer.

The triangle, drawn



Slide 56

Working through it

Where the triangle sits. Between the demand curve and the *social cost* curve, spanning the quantities from the optimum (30) to the equilibrium (40).

Its three corners, each of which means something.

- At $Q = 30$ the two curves cross, so the triangle closes to a point. This is the last bundle worth burning.
- At $Q = 40$ on the demand curve: \$6.00, the value of the last bundle actually burned.

- At $Q = 40$ on the social cost curve: \$8.00, what that bundle truly cost. The vertical side between these two is the \$2 of loss on the final bundle.

Why the area is the total loss. At each quantity between 30 and 40, the vertical gap between social cost and demand is the value destroyed by that bundle. Adding the gaps across the range gives the area, which is the triangle.

The shape to recognize. A triangle pinched between two curves, spanning the difference between the quantity that happens and the quantity that should, means the same thing everywhere it appears: trades at the wrong margin.

Make the market see the smoke

Make the market see the smoke

Internalizing an externality. Restructuring incentives so the people in the trade face the full social cost (or value) of it.

A **corrective tax** (a *Pigovian tax*) of \$2.00 per bundle — set equal to the external cost — charged to sellers:

$$Q^s : 10(P - 2) - 20 = 10P - 40 \quad 100 - 10P = 10P - 40 \Rightarrow P = \$7, \quad Q = 30$$

Buyers pay \$7; sellers keep \$5; the town collects $\$2 \times 3,000 = \$6,000/\text{week}$.

$Q = 30$: the taxed market lands *exactly* on the social optimum. This tax removes a deadweight loss.

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Working through it

The definition. **Internalizing an externality** means restructuring incentives so the parties to the trade face the full social cost or value of it.

The instrument. A **corrective tax** — also called a Pigovian tax — set equal to the external cost: \$2.00 per bundle, charged to sellers.

Working it through. Sellers now keep $P - 2$, so

$$Q^s = 10(P - 2) - 20 = 10P - 40$$

Setting demand equal to it: $100 - 10P = 10P - 40 \implies 140 = 20P \implies P = 7$, and $Q = 100 - 70 = 30$. Buyers pay \$7, sellers keep \$5, and the town collects $\$2 \times 3,000 = \$6,000$ a week.

Why $Q = 30$ exactly. The tax raises the curve sellers respond to by precisely the amount by which it fell short of social cost. The two curves now coincide, so the market's own equilibrium lands on the social optimum. The tax does not override the market; it corrects the price the market is using.

The twist worth savoring. A tax in a healthy market creates a deadweight loss by pushing quantity below the efficient level. Here it *removes* one, because this market was trading past the efficient level to begin with. Whether a tax helps or harms depends entirely on which side of the optimum the market was on.

7. Part 7: Benefits That Spill Over

Every ride is a car that isn't there

Every ride is a car that isn't there

E-bike share in a gridlocked city:

$$Q^d = 80 - 10P \qquad Q^s = 10P - 20$$

(Q in **hundreds** of rides per day.)

Equilibrium:

$$80 - 10P = 10P - 20 \Rightarrow 20P = 100 \Rightarrow P^* = \$5, \quad Q^* = 80 - 10(5) = 30$$

Every rush-hour ride is one car *not* in everyone else's way.

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Working through it

Flipping the sign. Part 6 dropped a cost on bystanders. This part drops a benefit, and the whole analysis runs in mirror image.

The market. E-bike share in a congested city, Q in **hundreds** of rides per day:

$$Q^d = 80 - 10P \qquad Q^s = 10P - 20$$

Solving.

$$80 - 10P = 10P - 20 \implies 100 = 20P \implies P^* = 5$$

and $Q^* = 80 - 10(5) = 30$, that is 3,000 rides a day. Check in supply: $10(5) - 20 = 30$. ✓

The spillover. Every rush-hour ride is one car *not* in the queue. The people who benefit are the drivers still on the road — who do not rent bikes, are not part of the transaction, and pay nothing toward it.

The symmetry to expect. A trade whose costs spill out gets over-produced, because the market undercounts its cost. A trade whose benefits spill out should therefore be *under*-produced, because the market undercounts its value. The next frames confirm it.

What a ride is truly worth

What a ride is truly worth

The bystanders' gain — less congestion, less exhaust, for drivers who never touch a bike: **\$2.00 per ride**.

Social value. Private value (WTP) plus external benefit: what one more unit is truly worth to society.

The rider weighs the fare against *their own* benefit only — the \$2 enters **nobody's** decision.

Mirror image of the smoke: a benefit the market never sees.

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Working through it

Pricing the spillover. Each ride is worth about \$2.00 to bystanders — congestion and exhaust that never happen — spread across the drivers who remain.

The definition. **Social value** is private value (the rider's WTP) plus external benefit: what one more unit is truly worth to society.

Where the blindness is. A would-be rider compares the \$5 fare with *their own* benefit from the ride. That is a sensible private decision. But the \$2 of value the ride hands to strangers in traffic enters nobody's calculation: the rider is not paid for it, the operator cannot bill for it, and the beneficiaries have no way to offer.

The exact mirror of the firewood case. There, a \$2 cost sat outside the transaction and the market treated it as zero, so too much was traded. Here, a \$2 benefit sits outside the transaction and the market treats it as zero, so — as the next frame shows — too little is.

The general statement. A market prices what is inside the transaction. Anything outside it is valued at zero by default, and the direction of the resulting error follows the sign of what was left out.

The rides that don't happen

The rides that don't happen			
Social value = $\frac{100-Q}{10}$; cost = $\frac{Q+20}{10}$:			
Rides (00s)	Social value	Cost	Society's gain
30	$\frac{100-30}{10} = 7.00$	$\frac{30+20}{10} = 5.00$	+2.00
35	$\frac{100-35}{10} = 6.50$	$\frac{35+20}{10} = 5.50$	+1.00
40	$\frac{100-40}{10} = 6.00$	$\frac{40+20}{10} = 6.00$	0.00
45	$\frac{100-45}{10} = 5.50$	$\frac{45+20}{10} = 6.50$	-1.00
Optimum: 4,000 rides. Market: 3,000. The missing thousand? The rider pays the whole fare; the bystanders' \$2 gain is priced at zero.			
With a positive externality, the market <i>undershoots</i> .			
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Working through it

Building the social value column. Inverting demand gives the rider's own value at quantity

Q as $\frac{80-Q}{10}$. Adding the \$2 external benefit:

$$\text{social value} = \frac{80-Q}{10} + 2 = \frac{100-Q}{10}$$

Cost, from inverting supply, is $\frac{Q+20}{10}$.

Reading the rows. At $Q = 30$ — where the market stops — social value is \$7.00 and cost is \$5.00, so society gains \$2.00 from one more ride. At $Q = 35$ the gain is \$1.00. At $Q = 40$ it is zero: the optimum. At $Q = 45$ it is negative.

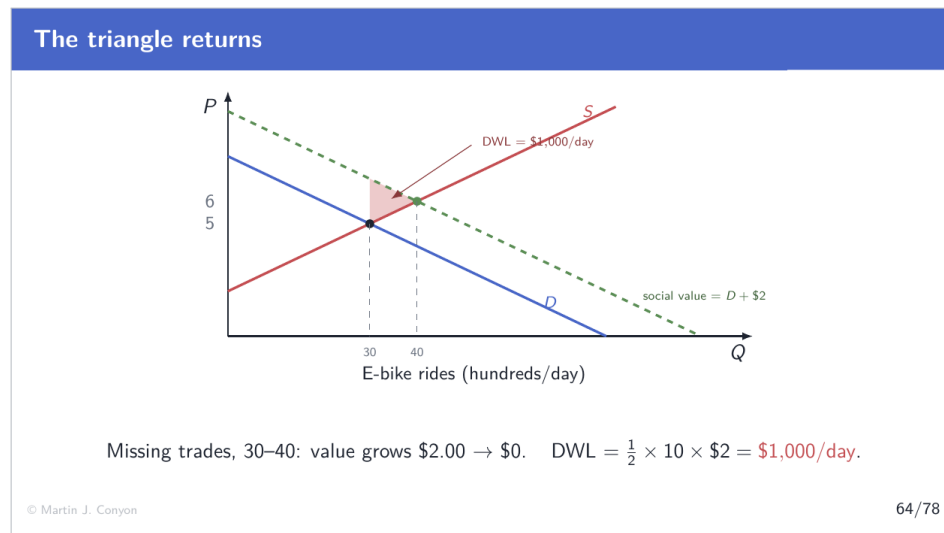
The optimum algebraically.

$$\frac{100-Q}{10} = \frac{Q+20}{10} \implies 100-Q = Q+20 \implies Q = 40$$

The shortfall. The optimum is 4,000 rides a day; the market delivers 3,000. A thousand daily rides that society values above their cost simply do not happen.

Why they do not. The marginal would-be rider values the ride at less than the fare, so declines — correctly, privately. The \$2 that ride would have handed to strangers is not part of their decision, and there is no mechanism by which it could be.

The triangle returns



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Working through it

Where the dashed line goes this time. Above demand. It is social value: the demand curve lifted by \$2 at every quantity. In Part 6 the dashed line sat above supply; the construction is the same, applied to the other curve.

Where the triangle sits. Between social value and supply, spanning the missing rides from $Q = 30$ (the market outcome) to $Q = 40$ (the optimum).

Computing it. At $Q = 30$ the gap between social value and cost is \$2.00. It narrows steadily to \$0.00 at $Q = 40$, where the curves cross. So

$$DWL = \frac{1}{2} \times 1,000 \times \$2.00 = \mathbf{\$1,000 \text{ per day}}$$

The same name, and the definition earns it. Part 6 named deadweight loss as surplus lost from trading the wrong quantity, *in either direction*. Here nothing is over-traded; rides that would have created value simply never occur. The value is destroyed by absence rather than by excess, and it is just as gone.

The visual rule. Whichever curve the market is failing to see, the dashed correction goes on that side, and the triangle opens between the true curve and the one the market is actually using.

Pay the spillover: a subsidy

Pay the spillover: a subsidy

Give riders **\$2.00 off** per ride — equal to the external benefit. Demand as sellers see it:

$$Q^d = 80 - 10(P - 2) = 100 - 10P \quad 100 - 10P = 10P - 20 \Rightarrow P = \$6, \quad Q = 40$$

Operators receive \$6; riders pay $6 - 2 = \$4$; city pays $\$2 \times 4,000 = \$8,000/\text{day}$.

The outlay is a **transfer** (taxpayers → riders/operators), not a loss. What society *gains*: the \$1,000/day triangle, back.

Tax what harms bystanders. Subsidize what helps them. Same logic, opposite signs.

Working through it

The mirror instrument. Part 6 taxed the harm. Here we *pay* the benefit: a subsidy of \$2.00 per ride, equal to the external gain.

Working it through. With a \$2 subsidy the rider's effective price is $P - 2$, so demand as the operators see it becomes

$$Q^d = 80 - 10(P - 2) = 100 - 10P$$

Setting it equal to supply: $100 - 10P = 10P - 20 \implies 120 = 20P \implies P = 6$, and $Q = 40$ — exactly the optimum. Operators receive \$6; riders pay $6 - 2 = \$4$; the city pays $\$2 \times 4,000 = \$8,000$ a day.

Keeping the accounting straight, because this is where it goes wrong. The \$8,000 is a **transfer** from taxpayers to this market — money moving between pockets, not value destroyed. It should not be counted as the cost of the policy in a surplus calculation.

What society actually gains. The \$1,000-a-day triangle of rides that now happen, and would not have. That is newly created value, not moved value.

The unifying rule for both parts. Make the price tell the truth: charge the harm, pay the help. The instrument differs only in sign.

The spillover scorecard

The spillover scorecard			
	Market vs. optimum	The triangle sits	Remedy
Negative ext.	trades too much	between D and social cost	tax = external cost
Positive ext.	trades too little	between social value and S	subsidy = external benefit

The market fails in the direction of whatever it cannot see.

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Working through it

Both cases on two lines.

Negative externality. The market undercounts cost, so it trades *too much*. The dead-weight loss triangle sits between demand and social cost, spanning optimum to equilibrium. The remedy is a tax equal to the external cost.

Positive externality. The market undercounts value, so it trades *too little*. The triangle sits between social value and supply, spanning equilibrium to optimum. The remedy is a subsidy equal to the external benefit.

The sentence that generates the whole table. **The market fails in the direction of whatever it cannot see.** Hide a cost and it overproduces; hide a benefit and it underproduces. You do not need to memorize the table — ask which side of the ledger is missing and the direction follows.

The right diagnosis of what is wrong. The market is not a bad calculator. Parts 1 to 5 established that it is an extraordinarily good one, arriving at the surplus-maximizing outcome with nobody in charge. It is a superb calculator being given incomplete inputs, and the repair is always to fix the inputs rather than to override the machine.

8. Part 8: Fixing It, With and Without Government

Two toolkits, and a third way

Two toolkits, and a third way

Command-and-control: forbid or require behavior directly — emission caps per firm, mandated equipment, permitted hours.

Market-based: put a price on the spillover and let people choose — corrective taxes and subsidies; *tradable permits*.

No government at all: sometimes the neighbors fix it themselves. (Last stop.)

The test for any tool: does it get the quantity right — at the lowest cost of getting it right?

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Working through it

Three families of remedy.

Command-and-control. Direct rules: emit no more than this, install that equipment, no leaf blowers before nine. The regulator specifies behavior.

Market-based. Put a price on the spillover and let people choose. The corrective tax of Part 6 was one instrument; tradable permits are the other.

No government at all. Sometimes the affected parties negotiate their own way to the efficient outcome. This is the last frame of the part, and it is less exotic than it sounds.

The single test to apply to all three. Does the tool reach the right quantity — *and* does it get there at the lowest possible cost? The second half of that test is what separates the families. A rule and a permit scheme can produce identical emissions and differ enormously in what achieving them consumes, and the next two frames show exactly that on numbers.

Why the distinction is not academic. The difference in the worked example is a factor of two, and it comes from information: a regulator writing a rule does not know what abatement costs each firm, and does not need to when the price does the sorting.

Two factories, one river

Two factories, one river

Northside Mill and Harbor Chemical each discharge **80 tons/month** into the same river. The town wants the total down to **100**.

Cost of cutting one ton: **Mill \$50 — Harbor \$150**.

Rule: “each firm cuts 30.”

	Tons cut	Cost
Northside Mill	30	$30 \times \$50 = \$1,500$
Harbor Chemical	30	$30 \times \$150 = \$4,500$
Total	60	\$6,000/month

The rule hits the quantity — but it is **blind to who can cut cheaply**.

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Working through it

The setup. Northside Mill and Harbor Chemical each discharge 80 tons a month, so 160 in total. The town wants the total down to 100, which means cutting 60 tons between them.

The asymmetry that matters. Cutting one ton costs the Mill \$50 and costs Harbor \$150. Harbor is three times as expensive to clean up.

The obvious rule: each firm cuts 30.

$$\text{Mill: } 30 \times \$50 = \$1,500 \quad \text{Harbor: } 30 \times \$150 = \$4,500$$

Total: **\$6,000 a month**, and the target is met exactly.

What is wrong with it. Nothing environmental — the river gets its 60 tons of relief. The problem is the bill. Three-quarters of the \$6,000 is spent at the expensive plant, doing work the cheap plant could have done for a third of the price.

Why the rule cannot do better. It is written in tons, not dollars, and the regulator who wrote it does not know the two firms’ abatement costs. Those numbers live inside the firms. A uniform rule is what you write when you cannot see them — and the next frame is a mechanism that does not need to.

Sell the right to pollute

Sell the right to pollute

Tradable pollution permits. Each permit licenses one ton of discharge. Total permits issued = the cap. Firms may buy and sell them.

Issue 100 permits, 50 to each firm. They trade at \$100:

	Emits	Cuts	Net cost
Mill (sells 30)	20	60	$60 \times 50 - 30 \times 100 = \0
Harbor (buys 30)	80	0	$30 \times 100 = \$3,000$
Total	100	60	\$3,000/month — half the rule.

Same river — but the **cutting migrated to the cheap cutter**. Society's real bill: the Mill's \$3,000 of abatement; the permit payments are transfers.

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Working through it

The instrument. **Tradable permits:** each permit licenses one ton of discharge, the number issued equals the cap, and firms may buy and sell them. Issue 100 permits, 50 to each firm, and suppose they trade at \$100.

The Mill's decision. Cutting costs it \$50 a ton; a permit is worth \$100. Selling a permit and cutting instead nets \$50 a ton, so it cuts as much as it can. It cuts 60 tons (cost $60 \times 50 = \$3,000$), emits 20, and sells its 30 spare permits for $30 \times 100 = \$3,000$. Net cost: **\$0**.

Harbor's decision. Cutting costs \$150 a ton; a permit costs \$100. Buying is cheaper. It cuts nothing, buys 30 permits for \$3,000, and emits 80.

Check the cap. $20 + 80 = 100$ tons. ✓ The river gets exactly what the rule delivered.

Society's real bill. The only real resources consumed are the Mill's \$3,000 of abatement. The \$3,000 of permit payments is a *transfer* between firms — it buys no cleanup and destroys nothing. So the true cost is \$3,000 against the rule's \$6,000: **half**.

The mechanism. No official ever learned that the Mill cleans for \$50 and Harbor for \$150. The permit price made it profitable for each firm to act on its own cost, and the cleanup migrated to the cheap cleaner by itself.

Tax or permits: price vs. quantity

Tax or permits: price vs. quantity

Both put a **price on a ton of harm**:

- **Corrective tax** — fixes the *price* of harming; quantity finds itself.
- **Permits** — fix the *quantity* of harm; the price finds itself.

Either beats a uniform rule when firms differ — and firms always differ.

Price the harm; let the market find who abates.

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Working through it

One idea, two implementations. Both instruments make harming bystanders cost money. Once it costs money, self-interest routes abatement to whoever can do it cheapest, which is exactly what the uniform rule could not achieve.

Where they differ.

- A **corrective tax** fixes the *price* of a ton of harm. How much total abatement results is then whatever firms choose — the quantity emerges.
- **Permits** fix the *quantity* of harm at the cap. What a ton ends up costing is then whatever the market clears at — the price emerges.

Why the choice is a genuine debate. If you care most about hitting an exact environmental target, permits guarantee it and let costs fall where they may. If you care most about not imposing unknown costs on industry, a tax fixes the cost and lets the quantity vary. Which risk you prefer to bear is a real policy question, not a technical one.

What is not in doubt. Either instrument beats a uniform rule whenever firms differ in abatement cost — and firms always differ. The rule's weakness is not its strictness; it is that it spends money where money buys the least cleanup.

No government: strike a deal

No government: strike a deal

Café Solana's Friday live-music night. The noise reaches Harriman Books next door.

worth to café: \$400/wk costs bookshop: \$600/wk

The café has the legal right to play.

The bookshop offers \$500 for quiet Fridays. Café accepts ($500 > 400$): music stops. Bookshop keeps $600 - 500 = \$100$; café gains $500 - 400 = \$100$.

Efficient outcome ($600 > 400$: quiet is worth more) — reached by a private check, no regulator in sight.

(If music were worth \$800 to the café: no offer the bookshop can afford tempts it. Music plays — *also* efficient.)

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Working through it

The situation. Cafe Solana's Friday live music is worth \$400 a week to the cafe. It costs Harriman Books next door \$600 a week in lost evening trade. The law gives the cafe the right to play.

Which outcome is efficient. Quiet is worth \$600 and music is worth \$400, so quiet creates more value. Efficiency requires the music to stop — and note that the law currently permits it.

The bargain. The bookshop offers \$500 for quiet Fridays. The cafe compares \$500 against the \$400 the music is worth to it, and accepts.

- The bookshop avoids \$600 of losses and pays \$500: it gains \$100.
- The cafe gives up \$400 of value and receives \$500: it gains \$100.

Both are better off, and the efficient outcome is reached with no regulator involved. The bystander stopped being a bystander by making an offer.

The variant, which is just as important. Suppose the music were worth \$800 to the cafe. Then no offer the bookshop can afford — it will not pay above \$600 — tempts the cafe,

so the music plays. That is *also* efficient: music worth \$800 beats quiet worth \$600. The bargaining does not favor quiet; it favors whichever use is worth more.

The Coase theorem — and its limits

The Coase theorem — and its limits

Coase theorem. If the affected parties can bargain at no cost, they reach the efficient outcome on their own — *whichever side the law initially favors*.

Flip the right (a quiet ordinance, music worth \$800): the *café* now pays (\$600–\$800 works) and plays on. **The right decides who pays whom, not what happens.**

Why bargaining fails at scale:

- transaction costs — lawyers cost more than the stakes;
- breakdowns — each side holds out for the larger slice;
- numbers — 5,000 households cannot bargain with every stove.

Two neighbors can Coase. A town cannot. That is when policy earns its keep.

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Working through it

The theorem. If the affected parties can bargain at no cost, they reach the efficient outcome on their own — *whichever side the law initially favors*.

Testing the “whichever side” clause. Flip the entitlement: a quiet ordinance, with music worth \$800 to the cafe. Now the cafe must buy permission, and any payment between \$600 (the bookshop’s loss) and \$800 (the cafe’s gain) leaves both better off. The music plays — the same outcome as when the cafe held the right.

What the legal right actually determines. Not what happens, but **who pays whom**. That is a distributional question, and often an important one, but it is separate from efficiency.

Why the world still needs corrective taxes. The premise — costless bargaining — fails routinely:

- **Transaction costs.** Lawyers and negotiation can cost more than the stakes.
- **Holdout and breakdown.** Each side has an incentive to stall for a larger share,

and deals that should happen collapse.

- **Numbers.** This is decisive. Five thousand valley households cannot each negotiate with every wood stove owner. The bargaining cost rises with the number of parties far faster than the stakes do.

The practical division. Two neighbors can bargain their way out. A town cannot — and that is precisely where a tax or a permit market earns its keep.